

SMA3262/SMA3272: ENGINEERING MATH 1B

Syllabus: Feb 2025-May 2025

Course Homepage: <https://somas2.uonbi.ac.ke/login/index.php>

The course homepage is on SOMAS: **SMA3262/SMA3272: Engineering Mathematics 1B**

All course information and resources will be available on this platform.

Instructor

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Lectures/ Tutorial Sessions

SMA3262: Wed 11-1 pm, Thu 9-11 am (Online, CB232A)

SMA3272: Tue 11-1 pm, Wed 9-11 am (Online, AW308)

Course Overview

The course is aimed at developing the basic mathematical skills for Engineering students that are imperative for effective understanding of Engineering subjects. The topics introduced will serve as basic tools for specialized studies in many Engineering fields. It consists of four main areas: Complex Numbers, Matrices and Linear equations, Ordinary Differential Equations (ODEs) and Vector Algebra/ Vector Calculus.

Prerequisite: SMA3261/SMA3271: Engineering Math 1A

Expected Learning Outcomes

At the end of successful completion of this course, the learner should:

1. Gain an understanding of the core ideas and concepts of the complex domain, linear algebra, ordinary differential equations and vector algebra
2. Demonstrate use of the complex domain in analysing given engineering problems
3. Solve systems of linear equations using matrix methods and linear algebra
4. Model given engineering problems as differential equations and solve them
5. Use vector algebra and calculus to solve problems involving vector quantities such as a system of forces
6. Apply rigorous and analytic approach to analyse and solve engineering problems using mathematical knowledge learnt in this course
7. Develop the mathematical maturity to undertake higher level courses in mathematics and engineering.

Student Learning Resources

- Lecture notes/ slides, video lectures, problem sets, past exams (available on SOMAS)
- Core Textbooks (pdfs available on SOMAS)
 1. Engineering Mathematics, K.A. Stroud, Dexter J. Booth. 8th Edition, Macmillan.
 2. Mathematics for Engineers, A. Croft, R. Davison. 4th Edition, Pearson.
- Additional Reference (pdfs available on SOMAS)
 1. Advanced Engineering Mathematics, Erwin Kreyszig. 10th Edition, John-Wiley and Sons Inc.
 2. Engineering Mathematics, John Bird. 4th Edition. Newnes.

Course Weekly Schedule

Week	Topics to be Covered	Reference from Core Text 1 (Stroud/ Booth)
1 17th Feb	- Course introduction. - Review of Complex Numbers	Part II, Programme 1 Pgs 379 - 391
2 24th Feb	- Complex numbers: geometrical representation of complex numbers, polar form, exponential form, De Moivre's theorem, roots of complex numbers. Applications	Prog 1 Pgs 393 - 402 Prog 2 Pgs 406 - 420
3 3rd Mar	- Matrices: elementary matrix operations, transposes, determinants, inverses.	Prog 4 Pgs 460 - 463, 475-476 Prog 5 Pgs 484 - 499
4 10th Mar	- Matrices: solution of a set of linear equations, Cramer's rule, inverse matrix method, Gaussian elimination method. Applications	Core text 2 (Croft/ Davison): Chap 13 Pgs 576 - 603 Prog 5 Pgs 499 - 502
5 17th Mar	- Ordinary differential equations (ODE's) Introduction to ODE's. Formation of ODE's. Solution of ODE's by direct integration, by separating variables, homogeneous equations – by substituting $y=Vx$.	Prog 25 Pgs 977 - 992
6 24th Mar	- Ordinary differential equations (ODE's) Linear equations - Solution of ODE's by use of integrating factor. Bernoulli's equation. Applications	Prog 25 Pgs 994 – 1007
7 31st Mar	- Ordinary differential equations (ODE's) Second order ODE's. Homogeneous equations. Applications	Prog 26 Pgs 1013 – 1021
8 7th Apr	- Ordinary differential equations (ODE's) Second order ODE's. Inhomogeneous equations. Applications	Prog 26 Pgs 1023 - 1033
9 14th Apr	Reading week -Review for mid- term CAT	- Mid-term CAT (face to face)
10 21st Apr	- Vector Algebra/ calculus Review of scalar and vector products Triple products	Prog 6 Pgs 530 – 543 Adv Eng Math (Stroud) Prog 22 pgs 771-784
11 28th Apr	- Vector Algebra/ calculus Derivatives of vectors Applications - Submission of group assignment	Adv Eng Math (Stroud) Prog 22 pgs 784-792
12 5th May	- Group assignment presentations	
13 12th May	- Reading Week - Final review	- End-material CAT (online)
14 19th May	- Read for final exam	
15 26th May	- Read for final exam	